CHEM 427/527 · SESSION 11 · THURSDAY, NOVEMBER 5, 2026

Filtering, smoothing, baseline correction: is this impurity above the reporting threshold?

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Smooth a noisy trace with a Savitzky-Golay filter, and choose its window from the narrowest peak. Show from the residual and the peak height that the window kept the signal.
2. Remove a drifting baseline with a polynomial through anchors that hold no peak. Read the anchor residual for a bow or a bump.
3. Report a small peak’s area as a percentage of the main peak’s area, and say whether it reaches the reporting threshold. State the baseline’s order and anchors.
4. Choose between two baseline orders from the anchor residual each leaves. Name the impurity reported under one order and not under the other.
5. Compare a coding assistant’s plan with your own data before approving it. Change the plan where your data show a feature that the plan does not account for.

FIRST, EVERYONE PULLS

Get session 11 with git pull

```
git pull
```

S11-preprocessing appears, with

- **data/chromatogram_a.csv**: Dataset A, the worked example
- **data/records.csv**: 30 records, one column each; each of you gets one
- **data/README.md**: the method's table of known peaks and how the files were made

Session 10 is due today

- Upload it by 1:30 pm.
- Fill in your prompt record, then Restart and Run All before you upload.

Nothing is downloaded today: both chromatogram files are in the repository.

FROM A WINDOW IN TIME TO A WINDOW ALONG A CHROMATOGRAM

What does a 40 s smoothing window do to a 10 s peak?

Session 10

A window multiplied the FID in time and broadened every line in the spectrum.

Today

A smoothing window slides along a chromatogram, point by point.

The question

What happens to a peak narrower than the window, and what survives?

Smoother, shorter, wider, or unchanged?

THE DECISION

Which impurities go into the batch record?

Who decides

A quality-control analyst reports every impurity at or above 0.05% of the main peak’s area, the ICH Q3A reporting threshold.

Called too low

An impurity that is growing from batch to batch is never reported, so there is no trend to read.

Called too high

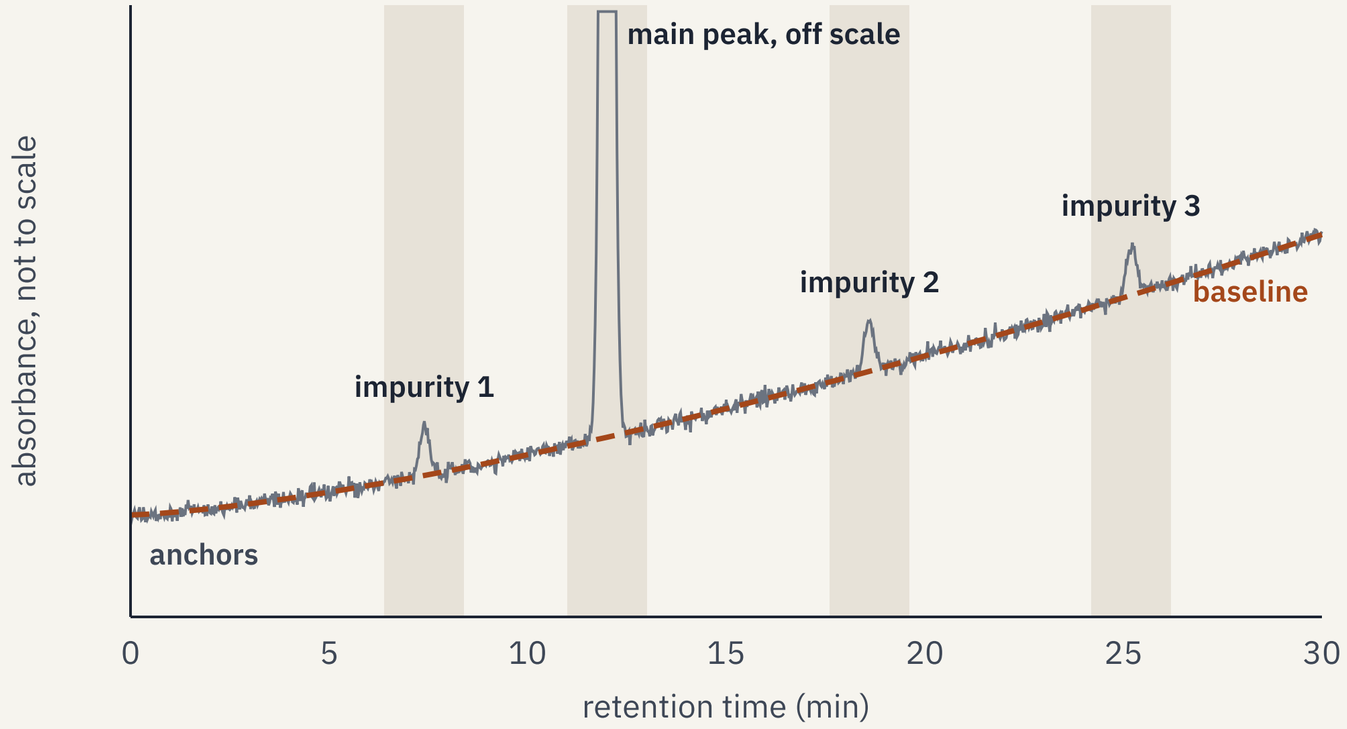
A stretch of baseline is reported as an impurity, and a batch is investigated for nothing.

Simulated: a gradient HPLC-UV chromatogram of a drug substance

30 minutes at 5 points per second, with a main peak near 1,200 mAU and three known impurities under 1 mAU. The baseline climbs 18 to 41 mAU through the run. Whether a small peak reaches 0.05% depends on how that baseline is removed.

THE METHOD'S TABLE AND YOUR FOUR DECISIONS

What you decide before you report



The method's table: the peaks the method expects

Peak	Retention time (min)
impurity 1	7.4
main peak	12.0
impurity 2	18.6
impurity 3	25.2

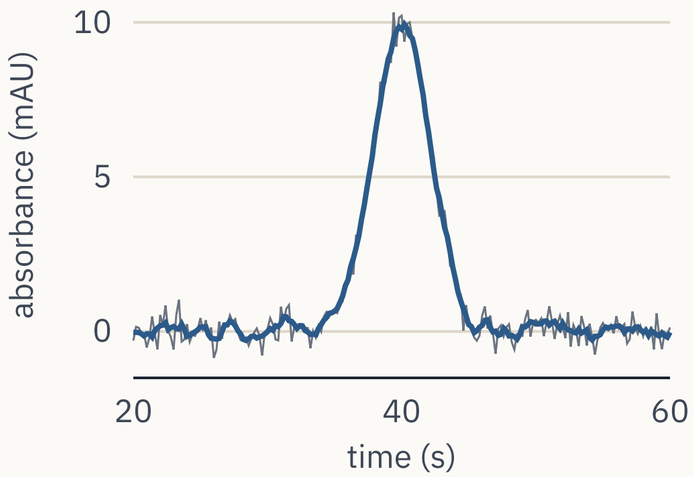
Your decisions

1. **Window**: how many points to smooth over (TASK 2)
2. **Anchors**: which stretches hold no peak (TASK 3)
3. **Order**: a straight line or a curve through the anchors (section 3)
4. **The call**: whether each impurity is at least 0.05% of the main peak's area

SMOOTHING, CHOICE 1: THE WIDTH OF THE WINDOW

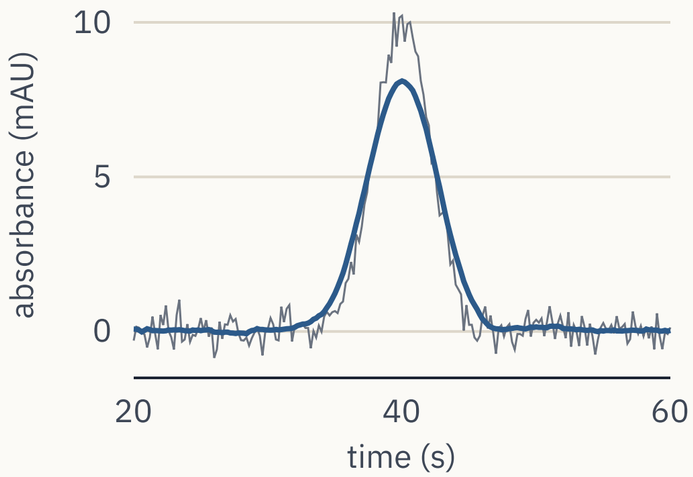
A rolling average at three widths

5 points, 1 s



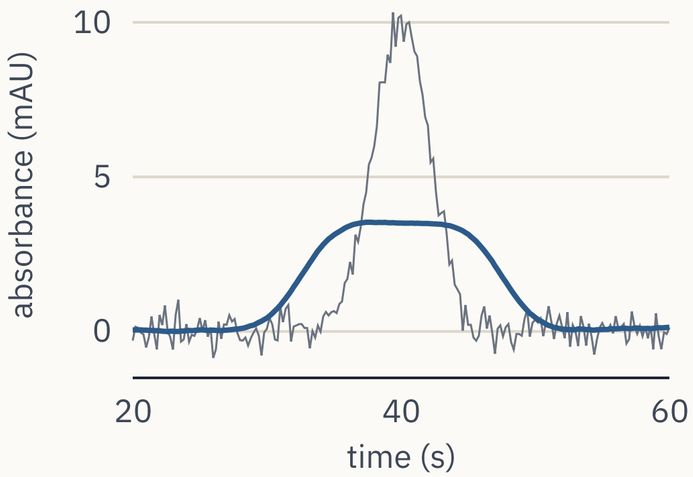
Height kept **99%**, noise **0.16** mAU

25 points, 5 s



Height kept **81%**, noise **0.07** mAU

75 points, 15 s



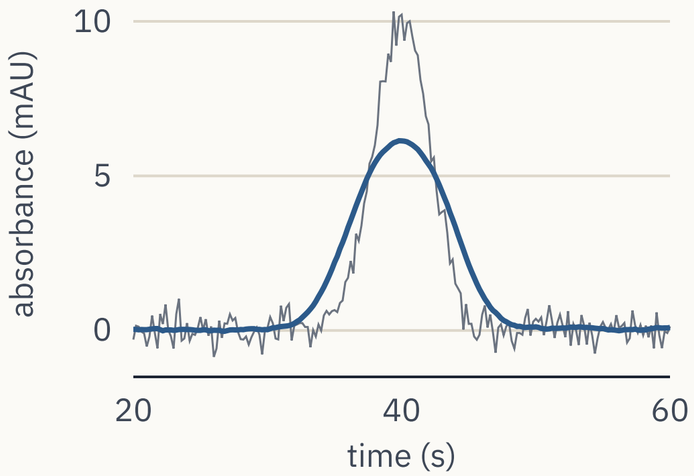
Height kept **35%**, noise **0.03** mAU

A wider window leaves less noise, from 0.37 mAU rms in the raw trace. It also flattens any peak narrower than the window.

SMOOTHING, CHOICE 2: THE ORDER OF THE POLYNOMIAL

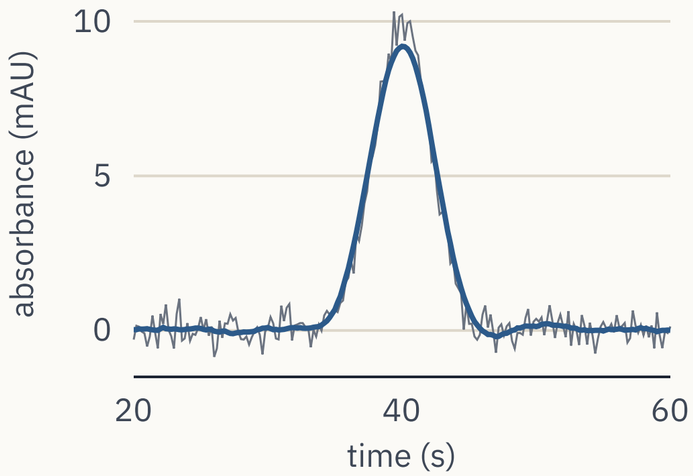
Savitzky-Golay, 41 points, three orders

Order 1, a rolling average



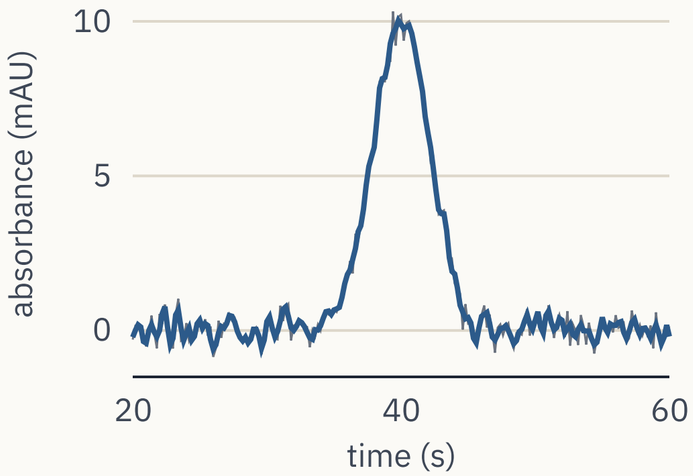
Height kept **61%**, noise **0.05** mAU

Order 3, as in the activity



Height kept **92%**, noise **0.07** mAU

Order 31, too far

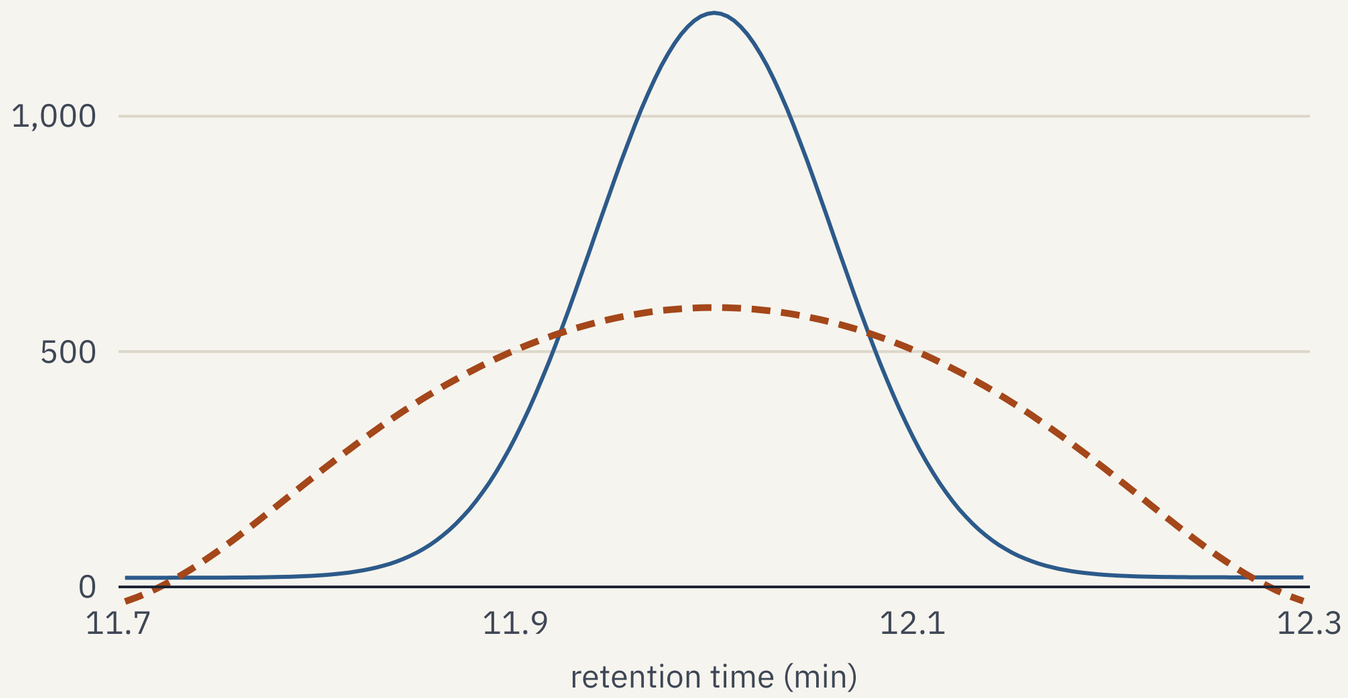


Height kept **100%**, noise **0.26** mAU

A higher order keeps more of a narrow peak’s height in the same window. Pushed too far, it follows the noise instead of the signal.

LIVE: THE MAIN PEAK UNDER TWO WINDOWS

Where did the rest of the peak go?



Raw: 1,220 mAU at 12.00 min

11 points, 2.2 s: 1,220 mAU, 100% kept

201 points, 40 s: 594 mAU, 49% kept

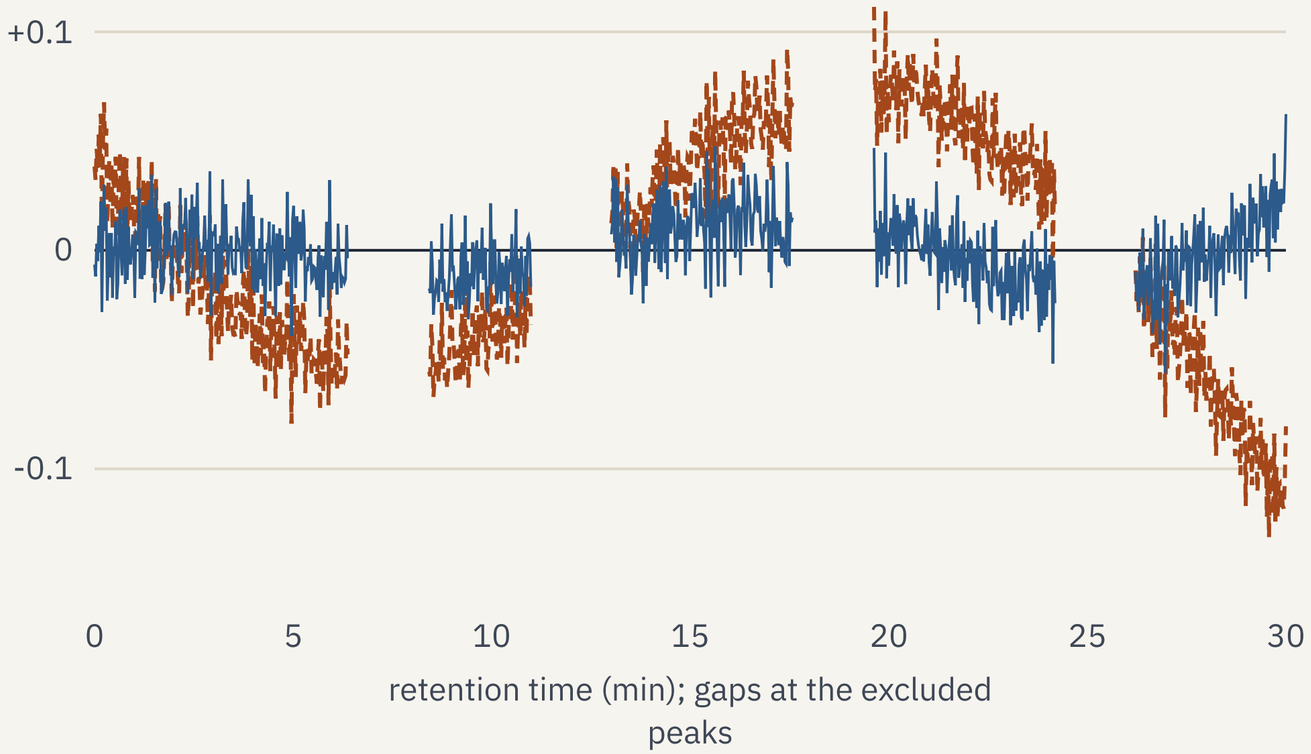
The residual, raw minus smoothed, inside the main peak: 0.05 mAU rms at 11 points, 334 mAU at 201, against 0.03 mAU of noise outside it.

The peak is 8.5 s wide at half height, 43 points. A window wider than the peak takes the peak’s shape into the residual.

The height goes. Does the area?

LIVE: THE ANCHOR RESIDUAL UNDER TWO BASELINE ORDERS

Which baseline do the data justify?



Order 1: a bow, down to 0.144 mAU

Order 3: noise within 0.063 mAU

The residual is the corrected trace inside the anchors, every point more than 1 min from a listed peak. It should be noise with no shape.

Impurity 2 at 18.6 min reads **0.065%** under order 1 and **0.039%** under order 3, across the 0.05% line. The bow is the difference.

The residual decides the order, and the order decides the call.

ASSIGNMENTS

One record each, by student number

#	Record	#	Record	#	Record	#	Record
1	record_01	7	record_07	13	record_13	19	record_19
2	record_02	8	record_08	14	record_14	20	record_20
3	record_03	9	record_09	15	record_15	21	record_21
4	record_04	10	record_10	16	record_16	22	record_22
5	record_05	11	record_11	17	record_17	23	record_23
6	record_06	12	record_12	18	record_18	24	record_24

Your record is your student number from session 4. In Parameters, set **RECORD**, such as RECORD = "record_07", run the Parameters cell, then the cell after TASK 1. Section 1 is the worked example on Dataset A. Section 2 is your record: the window, the plan, a first look, and the anchor cell you write. Section 3 fits both baseline orders through your anchors.

AI PRACTICE 4 OF 8, MET IN SESSION 7: EXPLORE AND PLAN BEFORE CODE

A plan in words, read against your record

Live, on a spare record, run through TASK 2 and saved

1. In Claude Code, click the mode indicator under the prompt box and choose Plan, or type /plan and a space.

2. Ask in your own words for a plan for the baseline step on this record in activity.ipynb: the anchors, the order, the window, the order of the steps. No code yet.

3. Read the plan’s anchor sentence aloud.

4. Run the first look, put its plot beside the plan, and ask: does the anchor sentence cover what this record shows?

What to read in a plan

- Where the anchors are, in words, against the peaks your record shows.
- The order, and what residual it expects.
- The window, against the width of the narrowest peak.

A plan built from the method’s table knows the table. Your record is the test of it.

SECTION 2: THE PLAN, THEN THE FIRST LOOK, THEN TASK 3

Your prompt, your check, your cell

Ask before you look. Save the notebook first.

1. After TASK 2, save, switch to Plan mode, and ask in your own words for the baseline plan on your record in activity.ipynb. Add the record cell and write your prompt on its first line.
2. Run the first look. Read the plan's anchor sentence against the plot and the printout.
3. Fill the record's other two lines: what the plan proposed and what the first look showed; what you changed, or verified.
4. TASK 3: write the anchor cell yourself, with the change in it.

The record, a markdown cell

```
**Prompt record**
- **My prompt:**
- **What the plan proposed for the anchors, and
what the first look showed about them:**
- **What I changed before approving, or what I
verified:**
```

TASK 3: a code cell you write that sets extra_t_min and anchors, by section 1's anchor_mask, excluding every peak your record shows.

The three lines of your prompt record count toward execution.

THE JUDGMENT CELL, SECTION 4

Your batch record, and one entry

(a) Your answer

- Which peaks of your record go into the batch record, and at what percent?
- The order you chose, with its anchor residual beside the other order’s
- The impurity reported under one order and not under the other
- The level of any peak from a degradation product or a contaminant
- What a wrong call costs either way

(b) One judgment entry

- Your prompt record is the source.
- What you did: changed, rejected, or verified
- The evidence: the anchor residual with the plan’s anchors against yours, and the time it sat at

Which of the two residual plots would you want filed with the batch record, and why?

SECTION 1 IS READ AND RUN. SECTION 2 IS YOURS.

Go, then submit

Submit

Fill in your prompt record, then Restart, Run All, save, and upload activity.ipynb by **1:30 pm Tuesday, November 10**.

Scoring, 0 to 3

1 if it runs after a restart and holds your prompt record, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas, or Chris in the room.

The third hour is an office hour.